

Isaac Mubiru

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PROFESSIONAL SUMMARY

Full-stack engineer (**React + Node.js**) with ~7 years building and operating production systems, and a track record of leading both teams and programmes. Led a **team of 4 developers** delivering legal-tech systems and architected their **microservices backend on Apache Kafka, Docker and Kubernetes**. At MTailor, owned the **end-to-end migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO — cutting **\$5,000/month** of infrastructure cost while the business kept trading. Ships product as well as platform: Express Shipping added **+\$40 revenue per order**. Now looking to set technical direction and raise the execution bar on developer platforms, dashboards and internal tooling. Code: github.com/zac-09

CORE COMPETENCIES

React & Node.js Full-Stack

Technical Leadership

API Design

Microservices Architecture

Docker & Kubernetes

Cloud Delivery (GCP / AWS)

Production Migrations & Delivery Health

Event-Driven Systems

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Own full-stack delivery across **React** front ends and **Node.js** back ends, reporting directly to the CTO with no intermediate layer.
- Shipped a 3D product-visualisation feature (video overlay + **ffmpeg**) that increased buyer conversion, and Express Shipping, generating **+\$40 revenue per order**.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime** on live traffic.
- Built **real-time two-way sync** between MongoDB and Firestore using Node.js and Google **Pub/Sub**, letting both stacks serve production traffic in parallel throughout cutover.
- Cut infrastructure cost by **\$5,000/month** by migrating all services off AWS; moved file storage from S3 to GCS via Python.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- **Led a team of 4 developers** building legal-tech systems for NGOs, setting technical direction and delivery standards across the stack.
- **Architected a microservices backend** for FIDA Uganda's case-management platform using **Apache Kafka, Docker and Kubernetes**.
- Designed a USSD legal-aid service communicating with a Node.js backend over **gRPC** — typed, schema-first service contracts.
- Delivered a cross-platform **React Native** + Expo app (iOS & Android) with Firebase push notifications, plus a paralegal database for LASPNET Uganda.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led **performance optimisation** of a PWA delivering real-time stock prices from a Firebase backend.
- Built jobs querying **2M+ Firestore records** against Morningstar APIs to keep customer-facing prices current.
- Added responsive **React** + Tailwind UI pages and extended backend **APIs** with Firebase Cloud Functions and Express.js.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Full-stack delivery for a US agency with demanding clients, reporting directly to the CTO — balancing **speed with quality** under tight timelines.
- Delivered **React** UIs from Figma designs in close partnership with design; built DeFi backends with Web3 and **Node.js**.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration pipeline reading from MongoDB and writing to Firestore after complex processing logic.

- Wrote engineering **documentation** for the new Firebase SDKs (Node.js & Python) and trained the Ops team on dashboards and backup procedures — operational readiness handover.

PROJECTS

Zero-Downtime Platform Migration — 20+ Applications

Programme-level ownership of a live migration from Parse/MongoDB to Firebase/GCP: parallel dual-write architecture, phased cutover, and an operational handover to the Ops team. Zero downtime; \$5,000/month saved.

Node.js · GCP Pub/Sub · Firestore · MongoDB · Python · AWS S3 → GCS

Microservices Backend — FIDA Uganda

Event-driven architecture for a legal case-management platform, designed and led alongside a 4-developer team, with a gRPC-backed USSD channel for feature-phone access.

Apache Kafka · Docker · Kubernetes · gRPC · Node.js · NATS Streaming

Healthcare WhatsApp Assistant

Applied-AI service integrating OpenAI and Google Vision for prescription OCR, delivered over the WhatsApp Business API.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Frontend: React, React Native, HTML, CSS, Tailwind **Backend:** Node.js, Express.js, REST APIs, gRPC, Microservices

Cloud / Infra: GCP, Firebase, AWS, Docker, Kubernetes, Linux **Data / Messaging:** MongoDB, Firestore, SQL, Google Pub/Sub, Apache Kafka

Languages: JavaScript, Python, Go (intermediate), Rust (intermediate) **Spoken:** English, Kiswahili